

Course Code: CSE3705	Course Name: Artificial Intelligence	
Credits: 3 (2-1-0)	Contact Hours: 2 hours Theory & 1 hour of discussion per week	
Batch: 2024 Academic Year: 2026-27	Semester Duration: 27th July to 20th November	
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Aim of the course: The course aims to provide students with a comprehensive understanding of Artificial Intelligence (AI), covering both foundational concepts and modern advancements in Generative AI. Students will learn how intelligent systems solve complex real-world problems through search, planning, reasoning, knowledge representation, and decision-making techniques. The course introduces key AI paradigms, including intelligent agents, expert systems, constraint satisfaction, and heuristic search methods. In addition, students will explore emerging AI technologies such as Generative AI, Transformer architectures, and Large Language Models (LLMs), gaining insights into how these systems generate content, process natural language, and support intelligent applications. The course also emphasises responsible AI practices, including ethical considerations, bias mitigation, privacy preservation, and trustworthy deployment of AI systems.

Course Overview and Context: Artificial Intelligence is a core course for Computer Science and Engineering students, providing the theoretical foundations and practical skills required to design intelligent systems. The course covers AI problem-solving through state-space search, planning, constraint satisfaction, knowledge representation, reasoning, intelligent agents, and expert systems. Building upon these fundamentals, students are introduced to Generative AI technologies, including Autoencoders, Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), Transformer architectures, and Large Language Models (LLMs). The course further explores model training, fine-tuning, evaluation methodologies, prompt engineering, data privacy, local LLM deployment, and application development using frameworks such as LangChain. Through hands-on exercises, demonstrations, case studies, and programming assignments using Python, students will gain practical experience in developing AI-enabled solutions and understanding the impact of modern AI technologies across diverse application domains. The course concludes with discussions on ethical and responsible AI, addressing issues such as fairness, bias, misinformation, transparency, and societal implications of AI systems.

Course Outcomes (CO): At the end of the course the students should be able to do the following:

CO Code	Course Outcome Statement	Mapped Component	Associated Activities / Assessment
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CO1	Understand theories and concepts necessary for building an artificial intelligence system.	Employability	Lectures, concept-based assignments, and MCQ-based quiz
CO2	Apply heuristics to develop a better searching algorithm, logic, and resolution to make logical inferences.	Skill Development	Lab exercises, MCQ-based quiz, and implementation assignment.
CO3	Analyze data to be able to deal with uncertainty by applying approximate reasoning and building simple intelligent agents.	Entrepreneurship, Skill Development	Hands-on projects, intelligent agent design, and development of working prototypes addressing real-world problems

Topics of the course:

S. No.	Topics	Number of Sessions	Course Outcomes
1	Introduction to Artificial Intelligence: Overview of AI concepts, terminology, application domains, AI vs. non-AI problems, Turing Test, evolution of AI, and contemporary AI applications.	4	CO1
2	Search and Problem Solving: State-space representation, uninformed search strategies, heuristic functions, A* algorithm, and optimality in search.	4	CO1, CO2
3	Planning and Constraint Satisfaction: Search versus planning, state-space planning, constraint satisfaction problems (CSPs), and applications of CSP techniques.	6	CO1, CO2
4	Knowledge Representation and Reasoning: Knowledge representation schemes, logical reasoning, inference mechanisms, and intelligent decision-making.	4	CO2
5	Intelligent Agents and Expert Systems: PEAS framework, types of intelligent agents, agent architectures, and components of expert systems.	6	CO1
6	Generative AI Fundamentals: Introduction to Generative AI, types of generative models, principles of data generation, applications, and model customisation for real-world use cases.	4	CO1, CO3
7	Generative AI Models: Autoencoders, Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), GAN architecture, training methodologies, and applications in image and video generation.	4	CO1
8	Transformer Architectures: Attention mechanisms, encoder-decoder architecture, self-attention, pre-training, fine-tuning, and applications of Transformer models.	8	CO1, CO3
9	Large Language Models and Responsible AI: Large Language Models (LLMs) and their role in understanding and generating human-like text; training datasets, training methodologies, evaluation metrics, fine-tuning approaches, local LLMs, data privacy, LangChain, prompt engineering, ethical AI, bias, misinformation, and responsible AI practices.	8	CO3
Total		48 Sessions	

CO/PO Mapping:

CO/PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	2	2	1	1								1	1		
CO2	2		1	2	2	2	1		1	1			1	2		
CO3			2	2										3		

Learning Resources:

Textbook:

1. Stuart Russel and Peter Norvig, **"Artificial Intelligence – A Modern Approach"**, Pearson.
2. Elaine Rich, Kevin Knight, and Shivshankar B Nair, **"Artificial Intelligence"**, McGraw Hill
3. Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play" by David Foster, O'Reilly Media. Second edition: June 6, 2023 (ISBN: 978-1098134181).
4. Generative AI with LangChain: Build large language model (LLM) apps with Python, ChatGPT, and other LLMs" by Ben Auffarth, Packt Publishing, first edition: December 22, 2023 (ISBN-13: 978-1835083468)
5. Build a Large Language Model (From Scratch)" by Sebastian Raschka, Manning Publications, 1st Edition, ISBN-13: 978-1633437166
6. Language Models: From Scratch to GPT-3" by Gabriel Garcia, Packt Publishing. ISBN: 978-1801815861

Reference Books:

1. George F. Luger, Artificial Intelligence: Structures and Strategies for Complex Problem Solving, Pearson
2. David L. Poole and Alan K. Mackworth, Python code for Artificial Intelligence: Foundations of Computational Agents, Cambridge University Press.
3. Vaswani, Ashish, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. "Attention is all you need." Advances in neural information processing systems 30 (2017). [Research Paper]
4. Programming PyTorch for Deep Learning – Ian Pointer, O'Reilly, 2020. Covers PyTorch basics, which is essential for implementing Transformers and generative AI models.
5. Yenduri, Gokul, Gautam Srivastava, Praveen Kumar Reddy Maddikunta, Rutvij H. Jhaveri, Weizheng Wang, Athanasios V. Vasilakos, and Thippa Reddy Gadekallu. "Generative Pre-trained Transformer: A Comprehensive Review on Enabling Technologies, Potential Applications.

Video lectures/ MOOC:

1. Stanford CS221: Artificial Intelligence: Principles and Techniques, By P. Liang & D. Sadigh
Video playlist: <https://www.youtube.com/playlist?list=PLoROMvody4rO1NB9TD4iUZ3qghGEGtgNX>
2. An Introduction to Artificial Intelligence, Prof. Mausam, IIT Delhi
Video Playlist: https://www.youtube.com/watch?v=K9gH7hBAp0&list=PLp6ek2hDcoNB_YJCruBFjhF79f5ZHyBuz&index=2
3. Coursera course: <https://www.coursera.org/learn/generative-ai-llm-architecture-data-preparation>

Note: Additional resources, wherever required, will be provided on a timely basis.

Experiential Learning Component:

The end-term project is the primary experiential learning component of this course. A student needs to spend approximately 15 to 20 hours to finish this assignment. The weightage of this component is 40%. The programming assignments (20% weightage) will help develop problem-solving abilities. Overall, the major focus of this course is on hands-on experiential learning. The total weightage to experiential learning is 60%.

Assessment Pattern: The final grade will be determined by the marks or grades earned during the project's phase-wise evaluations and the end-term assessment. Grading will be conducted using the relative grading method outlined in the university's academic regulations. To be eligible for grading, students must achieve a minimum of 40% of the total marks upon completing all assessments listed in the table below:

Evaluation Component	Weightage (%)	Evaluation Schedule	Remarks/Rubrics	
Assignment	20%	Second Week of September	The AI assessment will comprise both coding and viva components, designed to evaluate your practical and theoretical understanding of AI concepts. Both components are crucial, with the coding task emphasizing implementation skills and the viva focusing on your ability to articulate and explain your approach and AI concepts.	CO1, CO2
Midterm	20%	As scheduled by the University	Closed-book written exam.	CO1, CO2
Quiz	20%	Third week of November	The quiz will be conducted in MCQ format. Topics to be covered will be announced in classroom.	CO1, CO2
Project	40%	Project assignments will be provided one month before the end of the course. Evaluation will be conducted during the last two weeks of the course.	Assesses end-to-end AI project development: problem definition, literature review, data preparation, implementation, and reporting. Includes a 10% weightage pre-submission.	CO2, CO3
- All evaluations will be based on the work presented by the students as well as the questions asked, or the problems given to code. - Cases of AI-generated code or plagiarism will be taken seriously and reported according to the university's policy on Unfair Means (UFM). It is essential that all work is original and adheres to academic integrity. - There is a mandatory requirement to upload the project to a public repository on GitHub.				–

Student Responsibilities:

- Attend lectures and do the work Lab Assignments as per instructions.
- Participate in the discussions/assignments held during classes.
- Check announcements on the LMS and emails regularly.

- Submit the assigned task on time.
- Regularly check marks on the LMS to ensure they are up to date.
- Participate in class and take necessary actions to grasp the material. Asking questions is encouraged.
- Communicate any concerns by speaking directly with the instructor.

Attendance Policy: Students are expected to attend classes regularly. Failure to follow the classes regularly and adhere to the expected attendance percentage will result in losing quiz/lab marks and a reduction of the grade as per the University's grading policy.

Recourse Examination Policy: In case a student fails the course, a one-time recourse is permitted as per the academic regulations of the University. Recourse is allowed only for the End Semester examination.

Make-up policy: No make-up exam will be conducted for unexcused absences. The faculty needs to be informed in advance in case the student is not going to appear for any evaluation component, and it is at the discretion of the faculty to sanction makeup for an evaluation component.

Behavior Expectations: No mobile phones and other distractive gadgets are permitted in the class.

Academic Dishonesty/Cheating/Plagiarism: Plagiarism and dishonesty in any form in any evaluation component will lead to appropriate disciplinary action.